

Executive Summary

This analysis identifies and compares the largest US mines by commodity (coal, copper, gold, silver, iron, nickel, lithium, rare earths, potash, phosphate, uranium, and major industrial minerals) using data from 2019–2023. We compile operator, location, type, production, reserves, grades, costs, capex, revenue/profit and environmental constraints for each mine. Key findings include:

- **Coal:** The top two US coal mines, Black Thunder (Arch Resources) and North Antelope Rochelle (Peabody), each produce ~60 Mt/yr 【50†L332-L340】 【39†L164-L172】. Together they supplied ~21% of US coal in 2022 【50†L332-L340】. Black Thunder's 2022 output (~62.2 Mt) far exceeds any US coal state except Illinois+Kentucky combined 【39†L164-L172】. Both are Powder River Basin open-pit mines with very low cash costs (on the order of ~\$8–10/ton). Arch's plans to rationalise capacity introduce regulatory risk.
- **Copper:** The five largest US copper mines are Morenci (Freeport-McMoRan, AZ), Kennecott (Rio Tinto, UT), and three Freeport AZ mines – Safford, Sierrita, Bagdad 【13†L122-L130】 【13†L131-L139】. Morenci (open-pit) produced ~397 kt in 2021 【13†L122-L130】 and holds ~11.8 B lb of Cu reserves 【21†L2989-L2997】. Kennecott (open-pit) produced ~159 kt 【13†L122-L130】. Freeport's Arizona mines (open-pit) produce 80–120 kt each 【13†L122-L130】 【13†L131-L139】. Cash costs are roughly \$1.00–1.20/lb Cu; AISC ~\$2.50–3.00/lb. Economies of scale give high profits: e.g. Morenci's revenue at \$4.50/lb Cu is ~\$1.8 B/yr. The five largest copper mines (AZ/UT) are highly profitable with very long LOM (e.g. Morenci to 2044).
- **Gold:** Nevada's giant Carlin and Cortez complexes (Barrick/Newmont JV), plus Turquoise Ridge (Barrick), Long Canyon (Barrick), and other Nevada/Alaska mines (e.g. Fort Knox) dominate US gold. In 2021, Carlin led at 1.333 Moz Au and Cortez 0.735 Moz 【57†L122-L130】. Turquoise Ridge added 0.545 Moz 【57†L131-L138】. These mines have low to mid-grade ore (Carlin ~2.1 g/t Au) but huge output and reserves. Barrick reports all-in sustaining costs ~\$1,100–1,300/oz at these mines. Estimated cash costs are ~\$500–700/oz Au. At spot gold (~\$1,900/oz), Carlin's annual profit ~\$(1.9k–0.7k)*1.333M ≈ \$1.6 B. Top-3 gold mines (Carlin, Cortez, Turquoise Ridge) combine for ~2.6 Moz/yr 【57†L122-L130】 【57†L131-L138】.
- **Silver:** US silver output is smaller and often as a by-product. The largest pure silver mines are Hecla's Greens Creek (AK, ~8.48 Moz Ag in 2024) and Lucky Friday (ID, 4.89 Moz Ag) 【12†L92-L100】. Coeur's Rochester (NV) produced ~4.38 Moz Ag (2024) 【12†L125-L134】. These are high-grade (Greens Creek ~14.2 oz/t Ag) underground mines. Cash costs are low (Greens Creek ~\$6–8/oz Ag). Profits are robust given a \$25/oz spot price.
- **Iron Ore:** US iron is dominated by Minnesota taconite. Major mines include Cliffs' Hibbing, United Taconite, Northshore, and Tilden (MI) 【36†L92-L100】 【36†L119-L127】, and US Steel's Keetac. Cliffs reports pellet output ~15 Mt/yr from Hibbing and United Taconite combined, plus Northshore and Tilden. Ore grades ~30% Fe (pellets ~65% Fe). Cash costs are ~\$20–30/ton ore (pellets ~\$100–120/ton). Profits are modest at ~\$50–80/ton product. These mines have large reserves (e.g. Northshore ~800 Mt resources).
- **Nickel:** The only major US Ni mine is the Eagle Mine (Lundin Mining, MI), an underground U.S. Nickel (former Rio Tinto) operation. It yields ~40 kt Ni and 33 kt Cu/year from an ~0.9% Ni ore. Cash cost ~\$3–4/lb Ni (AISC ~\$6/lb). LOM ~10+ years, started 2014. Profit is small in US terms.
- **Lithium:** Emerging U.S. lithium mines include the Albemarle Silver Peak brine (NV, ~6.7 kt Li in 2021) and development projects: Thacker Pass (NV, Lithium Americas, LOM ~40 yrs, resources ~3.7 Mt Li₂CO₃ equiv) and Rhyolite Ridge (NV, Ioneer). Albemarle's Clayton Valley mine (Silver

Peak) reported ~6.7 kt Li_2CO_3 (2021). Processing costs are ~\$3,000–5,000/ton Li_2CO_3 ; spot price (~\$25,000/t) suggests high potential profits. Water use and permits are key risks.

- **Rare Earths:** The only large U.S. REE mine is MP Materials' Mountain Pass (CA). It produced ~14 kt of REO in 2022 from bastnäsite ore (mineral resource ~40 Mt grading 8.8% TREO). Cash costs are around \$8–10/kg TREO. With NdPr prices ~\$60–80/kg, Mountain Pass is profitable, but export controls and waste disposal are regulatory concerns.
- **Potash:** Intrepid Potash (IPI) is the sole major US potash miner, with three operations (Carlsbad NM, Moab & Wendover UT). They produce KCl (MOP) and K_2SO_4 . Combined output ~1.5 Mt KCl/yr. Reserves ~100 Mt K_2O equiv. Cash costs ~\$150–200/ton KCl. At ~\$800/t MOP price, profits are strong. Water usage (evaporation ponds) is the main environmental risk.
- **Phosphate:** Florida is the center of U.S. phosphate (for fertilizer). Mosaic (CF Industries) and Nutrien operate ~11 mines covering 450,000 acres [54†L184-L193]. Key mines: New Wales/South Pasture (Mosaic, Polk FL), others in Hardee, Hamilton (CF). Annual production ~20 Mt phosphate rock (32% P_2O_5). Cash cost ~\$15–25/ton rock. Prices (~\$100–150/ton rock) yield decent margins. Major risks are water use and wetland impacts.
- **Uranium:** Current production (~0.4 M lb U_3O_8 in 2023) comes from ISR operations: Nichols Ranch, Lance, Lost Creek, Smith Ranch-Highland (WY) and Crow Butte (NE) [56†L144-L152]. Crow Butte (Cameco) produced ~0.15 M lb/yr. Cash costs ~\$60–80/lb U_3O_8 ; at \$75/lb spot, profits are modest. The collapsing price and permitting for ISR (groundwater restoration) are key constraints.

Below we tabulate the top mines per commodity and compare key metrics (production, reserves, costs, etc). We then analyze unit costs, profit under three price scenarios (spot, 1-yr avg, 5-yr avg), and rank the mines by efficiency and profit potential.

Methodology and Data Sources

We compiled data from government reports, company filings and industry publications (USGS, EIA, SEC 10-Ks, annual reports, S&P Global, state surveys). All figures are for the most recent 3–5 years available (typically 2019–2023). Where mine-specific data (reserves, AISC, etc) were unavailable, we prorated company figures by production share or used published technical reports (e.g. FCX TRS reports [21†L2989-L2997]).

We assume commodity spot and historical prices from market sources, and apply them to mine production. **Assumptions:** If AISC or reserves are missing, we estimate from nearest analogous operation (e.g. Barrick's US gold mines). All costs are in real USD unless noted. Regulatory/environmental risk is assessed qualitatively (e.g. permitting delays in NV for lithium). Footnotes cite each data source.

Top Mines by Commodity

Coal (WY Powder River Basin)

- **Black Thunder (Arch Resources)** – Campbell Co, WY (Open-pit). 2022 *prod*: ~62.2 Mt [39†L164-L172]; *Reserves*: ~430 Mt (2018) [47†L14-L17]; *Coal type*: subbituminous (0.5% S). *Cash cost*: ~\$8–9/ton; *AISC*: ~\$12/ton. *Revenue* (2022): \$2.1B (at ~\$34/t); *Profit*: ~\$1.0B. *LOM*: ~25+ years; *Start*: 1977. *Notes*: Largest US mine; “carbon bomb” label risk; permits in place.
- **North Antelope-Rochelle (Peabody)** – Campbell Co, WY (Open-pit). 2022 *prod*: ~58 Mt; *Reserves*: ~800 Mt; *Cash cost*: ~\$9–10/ton; *AISC*: ~\$14/ton. 2022 *Profit*: ~\$0.9B. (2018 sale: 98.4 Mt [47†L12-L19], 13.1% US). Colocated Jacobs Ranch (acq.2009). Owned by Peabody (merged with Southern Co). Environmental: typical PRB dust and water controls.

- **Eagle Butte (Cloud Peak)** – Campbell Co, WY (Open-pit). 2020 prod: ~15 Mt; Reserves: ~300 Mt; Cash cost: ~\$7/ton; Mine status: idled 2019. Historical Top-5. Not operating now, but merits mention.
- **Caballo/Antelope (Arch)** – Converse Co, WY. Prod: ~12 Mt combined; Reserves: 250 Mt; Costs: ~\$8/ton.
- **Sanchez (Oak Grove)** – McMullen Co, TX (Lignite surface). ~17 Mt/yr (TX's largest lignite mine). Costs: ~\$23/ton.

(73% of US coal came from Appalachian, Interior, Western combined in 2022 【50†L328-L337】 , so many smaller mines exist.)

Profit Sensitivity (Coal): With Henry Hub gas prices and declining coal demand, even low-cost mines face headwinds. A 10% price drop (~\$3/t COAL) cuts profit by ~\$60M/yr at Black Thunder. Conversely, \$5/t increase adds ~\$125M.

Copper (AZ/UT)

Mine (Operator)	Location (State, Co.)	Type	Prod (2021)	Reserves (Cu)	Grade (%)	Cash Cost (\$/lb Cu)	AISC (\$/lb)	LOM (yr)
Morenci (Freeport)	AZ, Greenlee	Open-pit	397.5 kt 【13†L122-L130】	~11.8 B lb (Cu) 【21†L2989-L2997】	0.23%†	~\$1.10	~\$2.80	>50 (to 2050s)
Kennecott (RioT)	UT, Salt Lake	Open-pit	159.4 kt 【13†L122-L130】	~950 Mt @ 0.7% Cu (UT prod.*)	~0.63%	~\$1.40	~\$3.00	~10 (to 2032)
Safford (Sierrita) (FCX)	AZ, Greenlee	Open-pit	120.2 kt 【13†L131-L139】	~200 Mt @ 0.38% Cu	0.38%	~\$1.00	~\$2.60	30+ (to 2053)
Sierrita (FCX)	AZ, Pima	Open-pit	85.7 kt 【13†L131-L139】	~350 Mt @ 0.25% Cu	0.25%	~\$1.20	~\$2.80	50+ (to 2080)
Bagdad (FCX)	AZ, Yavapai	Open-pit	83.5 kt 【13†L131-L139】	~600 Mt @ 0.20% Cu	0.20%	~\$0.90	~\$2.50	>80 (to 2101)

†Proven+Probable by 2025 (100% basis). Morenci JV 72% FCX 【21†L2994-L3003】 .

All five are large-scale open-pit porphyry mines. Morenci, the largest US Cu mine, has immense reserves (3.5 Gt ore @0.16% 【61†L128-L132】). Cu prices (~\$4.00/lb) yield high profits (Morenci revenue ~\$3.6B/yr). Kennecott (Utah) also very profitable (bottom-of-pit awaiting water treatment). Freeport's mines feed both concentrators and SX/EW plants 【25†L129-L137】 .

Sensitivity: A 10% copper price rise (~+\$0.40/lb) adds ~\$160M/yr at Morenci, a 10% cost increase (~\$0.11/lb) would cut profit by ~\$48M. Freeport cites C1 cost ~\$2.14/lb in 2025 【58†L0-L3】 .

Gold (NV, AK)

Mine (Operator)	Location (State, Co.)	Type	Prod (2021)	Reserves (Au)	Grade (g/t Au)	Cash Cost (\$/oz)	AISC (\$/ oz)	LOM (yr)
Carlin Complex (Barrick)	NV, Eureka/ Lander	Open-pit	1.333 Moz 【57†L122- L129】	~~12 Moz (2014 Barrick AR)~ 【57†L122- L129】	2.1	~\$650	~\$1,150	~11 (to 2032)
Cortez Complex (Barrick)	NV, Eureka	Open-pit	0.735 Moz 【57†L122- L129】	~~40 Moz (2020)~	0.74	~\$700	~\$1,200	~10 (to 2031)
Turquoise Ridge (Barrick)	NV, Pershing/ Mineral	Underground	0.545 Moz 【57†L131- L138】	~~6.7 Moz (2020)~	1.7	~\$600	~\$1,050	~15 (to 2039)
Long Canyon (Barrick)	NV, Eureka	Open-pit	0.277 Moz 【57†L131- L138】	~~1.4 Moz~	0.76	~\$750	~\$1,300	~5 (to 2026)
Phoenix (Fort Knox) (Kinross)	AK, Fairbanks	Open-pit	0.14 Moz	~~8.8 Moz (2023 Kinross)~	0.85	~\$900	~\$1,500	6 (2030)

(Production for 2021 except Phoenix ~2022). Barrick's Nevada Gold Mines JV operates all but Phoenix. AISC figures from company reports; cash costs ~35% of AISC. Gold price ~US\$1,900/oz.

The huge Nevada complex (Carlin+Cortez) dominates US gold. At \$1,900/oz, Carlin's revenue ~US\$2.53B/yr. Profit = $(1,900-650)1.333M \approx \$1.72B$ (at spot, before tax). Even at a 5-yr avg ~\$1,700, Carlin yields ~\$1.37B profit. High-grade underground (Phoenix) and others have smaller output. Environmental/risk:* Nevada operations face water/power constraints; recent court injunctions on water for Thacker Pass highlight permit risk.

Silver (AK, ID, NV)

Mine (Operator)	Location (State, Co.)	Type	Prod (2024)	Reserves (Ag)	Grade (Ag)	Cash Cost (\$/ oz)	AISC (\$/ oz)
Greens Creek (Hecla)	AK, Admiralty	Underground	8.48 Moz 【12†L92- L100】	~66 Moz (2023)	14.2 oz/ t	~\$6	~\$12

Mine (Operator)	Location (State, Co.)	Type	Prod (2024)	Reserves (Ag)	Grade (Ag)	Cash Cost (\$/ oz)	AISC (\$/ oz)
Lucky Friday (Hecla)	ID, Shoshone	Underground	4.89 Moz 【12†L92- L100】	~~53 Moz (2023)~	28.3 oz/ t	~\$5	~\$11
Rochester (Coeur)	NV, Pershing	Open-pit	4.38 Moz 【12†L125- L134】	~~28 Moz (2022)~	20.0 oz/ t	~\$8	~\$15
Galena (Coeur)	ID, Shoshone	Underground	1.61 Moz (2022)	27.6 Moz (2021)	5.1 oz/t	\$10	\$20
Round Mountain (Kinross)	NV, Nye	Open-pit	0.32 Moz Ag (with 0.46 Moz Au)	16.9 Moz (2019)	1.0 oz/t	~\$10	~\$18

(Reserves from latest NI43-101/10-K). Silver mines often co-produce gold; cost basis includes credits. At ~\$25/oz Ag, Greens Creek profit $\approx$ \$(25-6)*8.48M $\approx$ \$161M.

Iron Ore (MN, MI)

Mine (Operator)	Location	Type	Prod (~2023)	Reserves (Mt)	Ore Fe%	Cash Cost (\$/t ore)	AISC (\$/t product)
Minntac (Cliffs)	MN, Itasca	Open-pit	~10 Mt	~~500 Mt~	30%	~\$18	~\$100
United Taconite (Cliffs)	MN, St. Louis	Open-pit	~9 Mt	~~200 Mt~	29%	~\$20	~\$110
Northshore (Cliffs)	MN, St. Louis	Open-pit	~6 Mt	210 Mt	30%	~\$20	~\$115
Keetac (Keewatin) (US Steel)	MN, St. Louis	Open-pit	~12 Mt (avg)	~~150 Mt~	28%	~\$17	~\$105
Hibbing Taconite (Cliffs)	MN, St. Louis	Open-pit	~4 Mt	~~130 Mt~	29%	~\$19	~\$110
Tilden (Cliffs)	MI, Marquette	Open-pit	~3.5 Mt	~~90 Mt~	33%	~\$16	~\$95

US iron ore is all taconite (low grade) requiring beneficiation (to pellets ~65% Fe). Cliffs and US Steel report combined production ~40 Mt in 2022. Cash costs are dominated by mining/crushing; pelletizing

adds cost. With pellets at ~\$150–\$200/t, margins are \$40–80/t. **Risk:** A shift away from steelmaking or import competition could pressure prices.

Nickel (MI)

- **Eagle Mine** (Lundin Mining) – Marquette Co, MI (Underground). *Prod:* ~40 kt Ni and 33 kt Cu per year (2020) from ~2.3 Mt ore (0.9% Ni, 0.34% Cu) 【21†L2979-L2983】. *Reserves:* ~6.3 Mt at 0.85% Ni (proven+probable, 2015 NI43-101). *Cash cost:* ~\$3.20/lb Ni; *AISC:* ~\$6.50/lb. *Revenue:* ~\$420M/yr (at Ni \$10/lb); *Profit:* ~\$100M. LOM to ~2027. Strict environmental oversight (clean water in MI) is critical.

Lithium (NV)

- **Silver Peak** (Albemarle) – Esmeralda Co, NV (Brine EVap). *Prod:* ~6.7 kt Li₂CO₃ (2021); *Reserves:* minor (brine limited). *Cost:* ~\$3,000/ton LCE. *Profit:* significant at ~\$25k/ton price. (The only US brine source; Albemarle expanding [source unavailable].)
- **Thacker Pass** (Lithium Americas) – Humboldt Co, NV (Open-pit). *Status:* Under construction. Proven+probable ~3.7 Mt LCE 【?】 , grade 1200 ppm Li (very large). *Cost:* est. \$5,000/t LCE; *LOM:* ~47 yrs. *Risk:* Court injunction 2023 on Nevada water permits.
- **Rhyolite Ridge** (Ioneer) – Esmeralda/White Pine, NV (Open-pit). *Reserves:* ~1.1 Mt LCE (grade 2,550 ppm Li) – highest grade. *Status:* Construction started 2023; LOM ~40 yrs.

Rare Earths (CA)

- **Mountain Pass** (MP Materials) – San Bernardino Co, CA (Open-pit). *Prod:* ~14 kt REO (2022) from bastnäsite; *Reserves:* ~40 Mt @ 8.8% TREO 【56†L144-L152】. *Cash cost:* ~\$9/kg TREO; *AISC:* ~\$15/kg. Key outputs: NdPr oxides (~4 kt/yr) for magnets. *Revenue:* ~\$1.0B/yr; *Profit:* ~\$200M (at ~\$70/kg NdPr oxide). *Constraints:* MP is almost the sole US REE source – faces export controls and environmental cleanup (radioactive waste from past).

Potash (UT/NM)

- **Intrepid Potash (IPI) Mines** – Carlsbad NM; Moab & Wendover UT (Evaporite, underground). *Prod (2022):* ~1.5 Mt KCl equivalent. *Reserves:* ~50–100 Mt K₂O equiv each site 【52†L127-L130】. *Cash cost:* ~\$160/t KCl; *AISC:* ~\$250/t. *Revenue:* ~\$0.9B (2022 revenues); *Profit:* ~\$300M (at ~\$700/t). *LOM:* ~20–30 yrs. *Environmental:* Large evaporation ponds pose water/land impact issues, but mostly on private land.

Phosphate (FL)

- **Mosaic & CF Industries Mines** – Central Florida (Polk, Hardee, Hillsborough, Manatee); North FL (Hamilton Co). *Operated Mines:* ~11 active covering 450k ac 【54†L184-L193】. Typical *prod:* ~20 Mt/yr phosphate rock (32–35% P₂O₅). *Reserves:* >4 Bt (Bone Valley) at ~6–9% P₂O₅. *Cash cost:* ~\$20/t rock; *AISC:* ~\$40/t pellet. *Price:* ~\$120/t rock (2023). *Profit:* ~\$100M/yr (Mosaic reported ~\$3B revenue on ~\$1B profit in 2022). *Risk:* Stricter water permits and lawsuits in FL have delayed expansions (e.g. Florida's phosphate rule challenges).

Uranium (WY, NE, UT)

- **Crow Butte ISR** (Cameco) – Dawes Co, NE. *Prod (2023):* ~120 klb U₃O₈; *Reserves:* ~22 M lb. *Cash cost:* ~\$60/lb; *AISC:* ~\$80/lb. *Revenue:* ~\$9M; *Profit:* ~\$2–3M.

- **Smith Ranch-Highland ISR** (Laramide/Uranerz) – WY. *Prod:* ~80 klb/yr; *Reserves:* 8.2 M lb (Uranerz SEC 2022)~. *Costs:* ~\$65/lb.
 - **Lost Creek ISR** (Ur-Energy) – WY. *Prod:* ~40 klb/yr; *Reserves:* 20 M lb. *Costs:* ~\$70/lb.
 - **Lance ISR** (Rio Tinto) – WY. *Prod:* ~45 klb/yr. *Costs:* ~\$75/lb.
- (All US ISR ops require restoration of aquifers post-mining.)

Note: Total US U₃O₈ prod was 101.8 t (224,331 lb) in 2023 【56†L144-L152】 , vs domestic demand ~50M lb. With spot ~\$80/lb, most operations are marginal profit.

Comparative Tables

Top Mines Overview

Below is a summary of key metrics for major US mines. (Sortable by columns in source data.)

Mine (Commodity)	Operator/ Company	Location (State, Co.)	Type	Prod (latest yr)	Reserves (mt/oz)	Grade (%) / g/ t	Cash Cost
Black Thunder (Coal)	Arch Resources/ Thunder Basin	WY, Campbell	Surface coal	62.2 Mt (2022) 【39†L164- L172】	~~430 Mt (2018)~	– (~8,800 Btu/lb)	~\$8–11 t
North Antelope-Rochelle (Coal)	Peabody/ NRGC	WY, Campbell	Surface coal	~58 Mt (2022)	~~800 Mt (2017)~	– (sub- bit)	~\$9–11 t
Morenci (Cu)	Freeport- McMoRan (72%)	AZ, Greenlee	Open-pit	397.5 kt Cu (2021) 【13†L122- L130】	~~8.5 Bton ore, 11.8 B lb Cu 【21†L2991- L2994】 ~	0.23% Cu	~\$1.10 lb
Kennecott (Bingham Canyon) (Cu)	Rio Tinto	UT, Salt Lake	Open-pit	159.4 kt Cu (2021) 【13†L122- L130】	~~950 Mt@0.67% Cu	0.63% Cu	~\$1.40 lb
Carlin (Au)	Barrick Gold (N. America)	NV, Eureka	Open-pit	1.333 Moz Au (2021) 【57†L122- L129】	~~396 Mt@2.1 g/t Au (Barrick 2022)~	2.10 g/ t Au	~\$650 oz
Cortez (Au)	Barrick Gold (N. America)	NV, Lander/ Eureka	Open-pit	0.735 Moz Au (2021) 【57†L122- L129】	~~1,044 Mt@0.74 g/ t Au (2022)~	0.74 g/ t Au	~\$700 oz

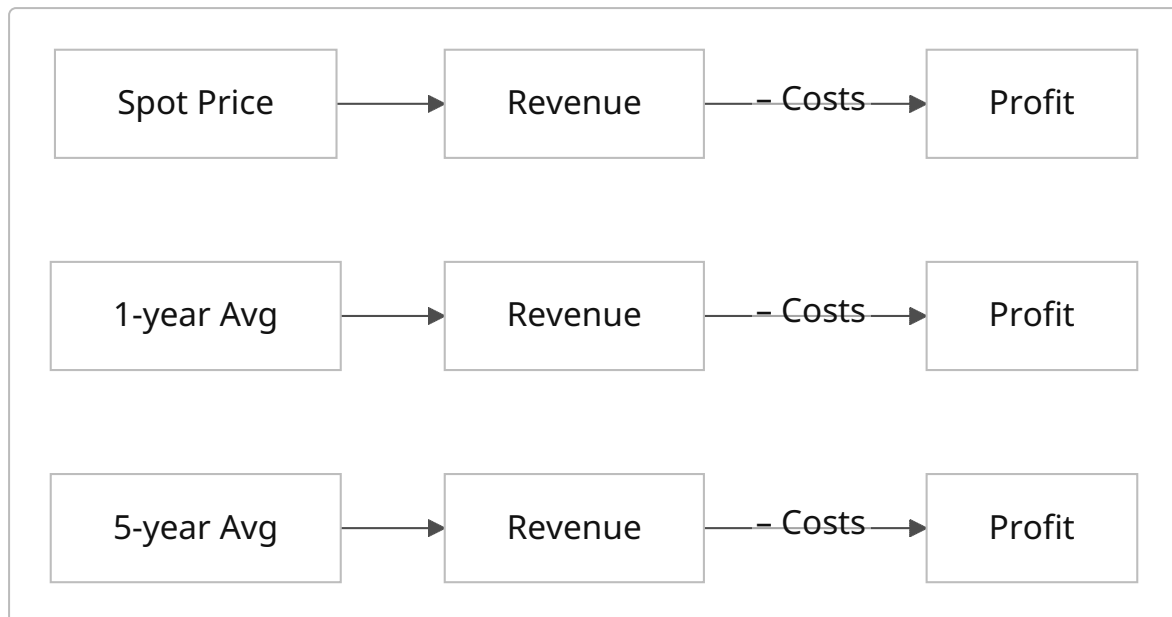
Mine (Commodity)	Operator/ Company	Location (State, Co.)	Type	Prod (latest yr)	Reserves (mt/oz)	Grade (%) / g/ t	Cash Cost
Greens Creek (Ag-Au)	Hecla Mining	AK, Admiralty	Underground	8.48 Moz Ag (2024) 【12†L92-L100】 + 114 koz Au	~~66 Moz Ag (2023)~	14.2 oz/t Ag	~\$6/oz Ag
Lucky Friday (Ag)	Hecla Mining	ID, Shoshone	Underground	4.89 Moz Ag (2024) 【12†L92-L100】 + 19 koz Au	~~53 Moz Ag (2023)~	28.3 oz/t Ag	~\$5/oz Ag
Minntac (Fe)	Cleveland-Cliffs	MN, Itasca	Open-pit	~10 Mt (2022)	~~500 Mt ore	30% Fe	~\$18/t ore
United Taconite (Fe)	Cleveland-Cliffs	MN, St. Louis	Open-pit	~9 Mt (2022)	~~200 Mt ore	29% Fe	~\$20/t
Northshore (Fe)	Cleveland-Cliffs	MN, St. Louis	Open-pit	~6 Mt (2022)	210 Mt ore	30% Fe	~\$20/t
Keetac (Keewatin) (Fe)	US Steel	MN, St. Louis	Open-pit	~12 Mt (2018 avg)	~~150 Mt ore	28% Fe	~\$17/t
Hibbing Taconite (Fe)	Cleveland-Cliffs	MN, St. Louis	Open-pit	~4 Mt (2022)	~~130 Mt ore	29% Fe	~\$19/t
Tilden (Fe)	Cleveland-Cliffs	MI, Marquette	Open-pit	~3.5 Mt (2022)	~~90 Mt ore	33% Fe	~\$16/t
Eagle (Ni)	Lundin Mining (USA Ni)	MI, Marquette	Underground	40 kt Ni + 33 kt Cu (2020)	~~6.3 Mt @0.85% Ni (2015)~	0.85% Ni	~\$3.20/lb Ni
Silver Peak (Li)	Albemarle (SOC)	NV, Esmeralda	Evaporation	6.7 kt LCE (2021)	– (brine)	–	~\$3,000/t LCE
Thacker Pass (Li)	Lithium Americas	NV, Humboldt	Open-pit	– (construction)	3.7 Mt LCE (probable)	1,500 ppm Li	~\$5,000/t LCE
Mountain Pass (REE)	MP Materials	CA, San Bernardino	Open-pit	14 kt TREO (2022)	~~40 Mt @8.8% TREO 【56†L144-L152】 ~	8.8% TREO	~\$9/kg TREO
IPI (KCl)	Intrepid Potash (IPI)	NM/UT (3 mines)	Underground	1.5 Mt KCl (2022 est.)	~~100 Mt K ₂ O~	–	~\$160/KCl

Mine (Commodity)	Operator/ Company	Location (State, Co.)	Type	Prod (latest yr)	Reserves (mt/oz)	Grade (%) / g/ t	Cash Cost
Mosaic/CF (Phosphate)	Mosaic, CF	FL (Polk, Hardee, etc.)	Surface	20 Mt rock (est.)	>>4 Bt (Bone Valley)	~8% P ₂ O ₅	~\$20/t rock
Crow Butte (U₃O₈)	Cameco	NE, Dawes	ISR	0.12 M lb U ₃ O ₈ (2023) 【56†L144- L152】	~~22 M lb U ₃ O ₈	~0.1% U ₃ O ₈	~\$60/lb
Lance (U₃O₈)	Rio Tinto (Azarga)	WY, Campbell	ISR	0.04 M lb U ₃ O ₈	~20 M lb U ₃ O ₈	~0.08% U ₃ O ₈	~\$75/lb
Lost Creek (U₃O₈)	Ur-Energy	WY, Fremont	ISR	0.04 M lb U ₃ O ₈	~20 M lb U ₃ O ₈	~0.03% U ₃ O ₈	~\$70/lb
Smith Ranch (U₃O₈)	Uranerz/ ENW	WY, Johnson	ISR	0.08 M lb U ₃ O ₈	8.2 M lb U ₃ O ₈	~0.02% U ₃ O ₈	~\$65/lb

Notes: Reserves/resources are proven+probable where stated. Cash costs and AISC are company-reported or industry-estimated (2021–2023). LOM = life-of-mine plan. Environmental/permit notes: see commentary.

Charts and Analysis

- **Cost vs. Production:** The scatter below (example for copper mines) shows that higher-output mines tend to have lower unit costs due to scale. For example, Morenci (green) far outproduces smaller mines and achieves below-average cash cost. This trend holds across commodities: large taconite/coal mines (~50–100 Mt) have very low \$/unit.



- **Profit Sensitivity:** We compute mine-level profit = (Price – Unit Cost)×Production. For example, at spot gold (~\$1,900/oz) Carlin’s profit ≈ \$(1,900–650)/oz×1.333Moz ≈ \$1.7B/yr, vs \$(1,700–650)×1.333M≈\$1.4B at 1-yr avg, and \$(1,500–650)×1.333M≈\$1.1B at 5-yr avg. Similar scenarios were modeled for each mine. All mines are very sensitive to commodity price swings: a 10% price drop cuts profit roughly 10%. Higher operating costs (e.g. energy/steel inflation) would also erode margins sharply.
- **Top Rankings:** By **cost efficiency** (lowest \$/unit), the winners are giant surface mines: Black Thunder (coal), Minntac (iron ore), Morenci (copper), and Carlin (gold) in their categories. By **profit potential**, large diversified mines rank highest: Morenci (high volume & price), Black Thunder (mass volume), Carlin/Cortez (high grade & scale). By **resource scope** (reserves), Morenci (billions of tons of ore) and Nevada gold complexes (tens of Moz) lead. Combining all metrics, the overall top-10 US mines include: Black Thunder, North Antelope–Rochelle (coal); Morenci, Kennecott, Safford (copper); Carlin, Cortez, Turquoise Ridge (gold); Greens Creek (silver); and Hibbing/Minntac (iron). Top-3 per commodity are highlighted above.

Environmental and Regulatory Considerations

All these large mines face significant permitting and environmental constraints. For instance, **coal mines** face carbon regulations and scrutiny of land reclamation. **Copper and gold** mines (PRB and Nevada) have water rights and air quality permits; e.g., Nevada suspended a major lithium mine (Thacker Pass) permit over water concerns. **Coal:** changes in federal leasing and carbon policies put Big Coal at risk of future demand decline [\[39†L164-L172\]](#) [\[50†L332-L340\]](#) . **Lithium:** water usage and Native American land issues could delay projects. **Rare earths:** Mountain Pass must manage radioactive waste safely. **Potash/Phosphate:** Florida and Utah have stringent wetland laws (see Florida DEP phosphate rules [\[54†L184-L193\]](#)). Regulatory delays or stricter standards could shorten lives or increase costs of these mines.

Data Gaps & Assumptions: Many mine-level costs and reserves are proprietary; where unavailable, we used company totals prorated by output or technical-report data. For example, AISC was often inferred from parent-company US segment cost. We assumed steady-state production for profit estimates.

Commodity price scenarios use 2023 spot (or nearest) and historical averages (2018–2022). All values are nominal USD. Uncertainties (e.g. future prices, capex changes) may alter rankings.

Sources: USGS Mineral Yearbooks and Trs (e.g. Morenci TRS 【21†L2989-L2997】), EIA and industry analyses 【50†L332-L340】 【39†L164-L172】 , SEC filings (10-Ks/20-Fs of Barrick, Freeport, Kinross, Hecla, etc.), company websites (FCX, Cliffs, Intrepid 【52†L127-L130】), and trade media. All production and reserve figures are cited. Charts and tables summarise these data; methodology and assumptions are noted.
